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**College of Engineering
& Applied Sciences**

LAB MANUAL

AY: 2026 – 27 (ODD SEMESTER)

SEMESTER: 3

DEPT(s) : CSE-ALLIED BRANCHES

SUBJECT NAME: OBJECT ORIENTED PROGRAMMING WITH JAVA

SUBJECT CODE: 1BCS302



OBJECT ORIENTED PROGRAMMING WITH JAVA LABORATORY

Subject Code : **1BCS302**

IAMarks: 50

Number of Lecture Hours/Week : 2T+ 2P

Exam Marks: 50

1. Develop a Java program to add TWO matrices of suitable order N.
2. Develop a stack class to hold a maximum of 10 integers with suitable methods. Develop a Java main method to illustrate Stack operations.
3. Create a Java class called Student with the USN, Name, Branch, Phone details as variables within it. Write a Java program to create N Student objects and print the USN, Name, Branch, and Phone of these objects with suitable headings.
4. Develop a Java program to create a class named Shape. Create three sub classes, namely: circle, triangle and square. Each class has two member functions named draw and erase. Demonstrate the concepts of polymorphism by developing appropriate methods, defining member data, and implementing the main program.
5. Develop a Java program to create an interface Resizable with methods `resizeWidth(int width)` and `resizeHeight(int height)` that allow an object to be resized. Create a class Rectangle that implements the Resizable interface and implements the resize methods.
6. Develop a Java program to create a package named mypack and import & implement it in a suitable class.
7. Develop a Java program to raise a custom exception (user-defined exception) for DivisionByZero using try, catch, throw and finally.
8. Develop a program to create a class MyThread with a class constructor, call the base class constructor, using `super` and start the thread. The run method of the class starts after this. It can be observed that both main thread and created child thread are executed concurrently.
9. Develop a Java program using an enumeration called Department containing CSE, ISE, ECE, and MECH. Accept a department name as input and display the selected department using `valueOf()`.
10. Develop a Java program to read the marks of three subjects as strings, convert them into wrapper objects, calculate the total and average using autoboxing/unboxing and display the result.
11. **Micro Project:** A team of two to three students shall develop a project and submit a brief report for a specific task assigned by the course instructor based on the concepts covered in the syllabus.

1. Develop a JAVA program to add TWO matrices of suitable order N (The value of N should be read from command line arguments).

Add.java

```
import java.util.Scanner;
public class Add
{
    public static void main(String[] args)
    {
        if (args.length != 1)
        {
            System.out.println("Usage: java MatrixAddition <order_M_N>");
            return;
        }
        int m, n; //Declare matrix size
        Scanner scan = new Scanner(System.in);
        m = Integer.parseInt(args[0]); //Initialize first matrix size
        n = Integer.parseInt(args[1]); //Initialize second matrix size

        int a[][] = new int[m][n]; //Declare first matrix
        int b[][] = new int[m][n]; //Declare second matrix
        int c[][] = new int[m][n]; //Declare third matrix
        //Initialize the first matrix
        System.out.println("Enter all the elements of first matrix:");
        for (int i = 0; i < m; i++)
        {
            for (int j = 0; j < n; j++)
            {
                a[i][j] = scan.nextInt();
            }
        }
        System.out.println("");
        //Initialize the second matrix
        System.out.println("Enter all the elements of second matrix:");
        for (int i = 0; i < m; i++)
        {
            for (int j = 0; j < n; j++)
            {
                b[i][j] = scan.nextInt();
            }
        }
    }
}
```

```
//Loop to add matrix elements
for (int i = 0; i < m; i++)
{
    for (int j = 0; j < n; j++)
    {

        c[i][j] = a[i][j] + b[i][j];

    }
}
//Print the resultant matrix
System.out.println("Matrix after addition:");
for (int i = 0; i < m; i++)
{
    for (int j = 0; j < n; j++)
    {
        System.out.print(c[i][j]+" ");
    }
    System.out.println("");
}
}
```

OUTPUT:

Java Add 3 3

Enter all the elements of first matrix:

1 2 3

4 5 6

7 8 9

Enter all the elements of second matrix:

9 8 7

6 5 4

3 2 1

Matrix after addition:

10 10 10

10 10 10

10 10 10

2. Develop a stack class to hold a maximum of 10 integers with suitable methods. Develop a JAVA main method to illustrate Stack operations.

Stack.java

```
import java.util.*;
public class Stack
{
    int s[]=new int[10]; int top= -1;
    int size=3;

    void push(int i)
    {
        if(top==size-1)
            System.out.println("Stack Overflow");

        else
        {
            s[++top] = i;
        }
    }
    void pop( )
    {
        if (top == -1)
        {
            System.out.println("Stack Underflow");
        }
        else
        {
            System.out.println(" Popped Element= " + s[top]);
            top--;
        }
    }
    void display( )
    {
        if(top == -1)
        {
            System.out.println("Stack is Empty\n");
        }
        else
        {
            System.out.println("Stack Elements are:\n");
            for (int i = top; i >= 0; i--)
                System.out.println(s[i]);
        }
    }
}
```

```
public static void main(String args[])
{
    Scanner scan = new Scanner(System.in);
    Stack stk = new Stack();
    for(;;)
    {
        System.out.println("\n---Stack Operations---");
        System.out.println("1. Push");
        System.out.println("2. Pop");
        System.out.println("3. Display");
        System.out.println("4. Exit");

        System.out.println("Enter your choice:\n");
        int choice = scan.nextInt();
        switch (choice)
        {
            case 1 :
                System.out.println("Enter the element
to push");
                stk.push(scan.nextInt());
                break;

            case 2 :   stk.pop();
                break;

            case 3 :   stk.display();
                break;

            case 4 :   System.exit(0);

            default :
                System.out.println("Invalid Choice\n");
                break;
        }
    }
}
```

OUTPUT:

-----Stack Operations-----

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

Stack is Empty

-----Stack Operations-----

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 2

Stack Underflow

-----Stack Operations-----

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the element to push 10

-----Stack Operations-----

1. Push
2. Pop
3. Display

4. Exit

Enter your choice: 1

Enter the element to push 60

-----Stack Operations-----

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the element to push 30

-----Stack Operations-----

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the element to push 50

Stack Overflow

-----Stack Operations-----

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

Stack Elements are:

30 60 10

-----Stack Operations-----

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 2

Popped Element= 30

-----Stack Operations-----

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 3

Stack Elements are:

60 10

-----Stack Operations-----

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 4

3. Create a Java class called Student with the USN, Name, Branch, Phone details as variables within it. Write a Java program to create N Student objects and print the USN, Name, Branch, and Phone of these objects with suitable headings.

```
import java.util.*;

public class Student
{
    String usn, name, branch;
    long phone;

    void insertStudent(String reg, String nm, String br, long ph)
    {
        usn=reg;
        name=nm;
        branch=br;
        phone=ph;
    }

    void displayStudent()
    {
        System.out.println("*****");
        System.out.println("USN= "+usn);
        System.out.println("NAME= "+name);
        System.out.println("BRANCH= "+branch);
        System.out.println("PHONE NUMBER= "+phone);
        System.out.println("*****");
    }

    public static void main(String args[])
    {
        Student st[]=new Student [100];
        Scanner ip=new Scanner(System.in);
        System.out.println("Enter the number of students");
        int n=ip.nextInt();
        for(int i=0;i<n;i++)
            st[i]=new Student();

        for(int j=0;j<n;j++)
        {
            System.out.println("Enter the Usn, Name, Branch, Phone
Number");
            String usn=ip.next();
            String name=ip.next();
            String branch=ip.next();
            long phone=ip.nextLong();
            st[j].insertStudent(usn, name, branch, phone);
        }

        for( int m=0;m<n;m++)
```

```
{  
    System.out.format("Student %d details are\n",m+1);  
    st[m].displayStudent();  
}  
}
```

OUTPUT:

Enter number of students: 2

Enter details of Student 1

USN: 101

Name: Ravi

Branch: CSE

Phone: 9876543210

Enter details of Student 2

USN: 102

Name: Anu

Branch: ECE

Phone: 9876543211

USN Name Branch Phone

101 Ravi CSE 9876543210

102 Anu ECE 9876543211

4.Develop a JAVA program to create a class named shape. Create three sub classes namely: circle, triangle and square, each class has two member functions named draw () and erase (). Demonstrate polymorphism concepts by developing suitable methods, defining member data and main program.

Shape.java

```
public class Shape
{
    // Member functions
    public void draw()
    {
        System.out.println("Drawing a shape");
    }

    public void erase()
    {
        System.out.println("Erasing a shape");
    }
}

// Circle class, a subclass of Shape
class Circle extends Shape
{
    public void draw()
    {
        System.out.println("Drawing a circle");
    }
    public void erase()
    {
        System.out.println("Erasing a circle");
    }
}

// Triangle class, a subclass of Shape
class Triangle extends Shape
{
    public void draw()
    {
        System.out.println("Drawing a triangle");
    }

    public void erase()
    {
        System.out.println("Erasing a triangle");
    }
}

// Square class, a subclass of Shape
class Square extends Shape
{
    public void draw()
    {
        System.out.println("Drawing a square");
    }
}
```

```
}

public void erase()
{
    System.out.println("Erasing a square");
}

// Main program to demonstrate polymorphism
class Main
{
    public static void main(String[] args)
    {
        // Creating objects of different shapes
        Circle c = new Circle();
        Triangle t = new Triangle();
        Square s = new Square();

        // Demonstrating polymorphism by calling draw and erase methods
        System.out.println("Using Circle object:");
        c.draw();
        c.erase();

        System.out.println("\nUsing Triangle object:");
        t.draw();
        t.erase();

        System.out.println("\nUsing Square object:");
        s.draw();
        s.erase();
    }
}
```



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OUTPUT:

Using Circle object:

Drawing a circle

Erasing a circle

Using Triangle object:

Drawing a triangle

Erasing a triangle

Using Square object:

Drawing a square

Erasing a square

5. Develop a JAVA program to create an interface Resizable with methods `resizeWidth(int width)` and `resizeHeight(int height)` that allow an object to be resized. Create a class `Rectangle` that implements the `Resizable` interface and implements the resize methods.

Rectangle.java

```
interface Resizable
{
    void resizeWidth(int width);
    void resizeHeight(int height);
}

class Rectangle implements Resizable
{
    private int width;
    private int height;

    public Rectangle(int width, int height)
    {
        this.width = width;
        this.height = height;
    }

    public void resizeWidth(int newWidth)
    {
        this.width = newWidth;
    }

    public void resizeHeight(int newHeight)
    {
        this.height = newHeight;
    }

    public void display()
    {
        System.out.println("Rectangle width: " + width + ", height: " + height);
    }

    public static void main(String[] args)
    {
        Rectangle rectangle = new Rectangle(5, 10);
        rectangle.display();

        rectangle.resizeWidth(8);
        rectangle.resizeHeight(15);
        rectangle.display();
    }
}
```



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Output:

Rectangle width: 5, height: 10

Rectangle width: 8, height: 15

6. Develop a JAVA program to create a package named mypack and import & implement it in a suitable class.

Package: mypack
Class name: MyClass

```
package mypack;  
  
public class MyClass  
{  
    public void display()  
    {  
        System.out.println("This is a method from the mypack package!");  
    }  
}
```

Example.java

```
import mypack.MyClass;  
public class Example  
{  
    public static void main(String[] args)  
    {  
        MyClass obj = new MyClass();  
        obj.display(); // Access the method from the imported package  
    }  
}
```

Output: Run Example.java

This is a method from the mypack package!

7. Develop a JAVA program to raise a custom exception (user defined exception) for DivisionByZero using try, catch, throw and finally.

CustomDivision.java

```
public class CustomDivision
{
    public static void main(String[] args) {
        int numerator = 10;
        int denominator = 0;

        try
        {
            if (denominator == 0)
            {
                throw new DivisionByZeroException("Division by zero is not allowed!");
            }
            int result = numerator / denominator;
            System.out.println("Result: " + result);
        }

        catch (DivisionByZeroException e)
        {
            System.err.println("Error: " + e.getMessage());
        }

        finally
        {
            System.out.println("This block always executes, regardless of exceptions.");
        }
    }
}

// Custom exception class for DivisionByZero
class DivisionByZeroException extends Exception
{
    public DivisionByZeroException(String message)
    {
        super(message);
    }
}
```

OUTPUT:

This block always executes, regardless of exceptions.
Error: Division by zero is not allowed!

8. Develop a program to create a class `MyThread` in this class a constructor, call the base class constructor, using `super` and start the thread. The `run` method of the class starts after this. It can be observed that both main thread and created child thread are executed concurrently.

TestMyThread.java

```
public class TestMyThread
{
    public static void main(String args[])
    {
        new MyThread();
        try
        {
            for( int k = 5; k > 0; k--)
            {
                System.out.println ("Running main thread :" + k);
                Thread.sleep(1000);
            }
        }
        catch (InterruptedException e)
        {
            System.out.println ("Exiting main thread . . .");
        }
    }
}

class MyThread extends Thread
{
    MyThread()
    {
        super("Using Thread class");
        System.out.println("Child thread:" + this);
        start();
    }
    public void run()
    {
        try
        {
            for ( int i =5; i > 0; i--)
            {
                System.out.println("Child thread" + i);
                Thread.sleep (500);
            }
        }
        catch (InterruptedException e)
        {
            System.out.println("exiting child thread ...");
        }
    }
}
```

```
}  
}
```

Output:

```
Child thread:Thread[Using Thread class,5,main]  
Running main thread :5  
Child thread5  
Child thread4  
Child thread3  
Running main thread :4 Child  
thread2  
Running main thread :3 Child  
thread1  
exiting child thread ...  
Running main thread :2  
Running main thread :1  
Exiting main thread ...
```

9. Develop a Java program using an enumeration called Department containing CSE, ISE, ECE, and MECH. Accept a department name as input and display the selected department using `valueOf()`.

```
enum Department {  
    CSE, ISE, ECE, MECH  
}  
  
public class DepartmentDemo {  
    public static void main(String[] args) {  
  
        Department dept = Department.valueOf(args[0]);  
  
        System.out.println("Selected Department: " + dept);  
    }  
}
```

OUTPUT:

Output

```
Selected Department: CSE
```

For example, if you run:

```
java DepartmentDemo ECE
```

The output will be:

```
Selected Department: ECE
```

10. Develop a Java program to read the marks of three subjects as strings, convert them into wrapper objects, calculate the total

and average using
autoboxing/unboxing and
display the result.

```
public class MarksDemo {  
    public static void main(String[] args) {  
  
        // Read marks as strings and convert to wrapper objects  
        Integer mark1 = Integer.valueOf(args[0]);  
        Integer mark2 = Integer.valueOf(args[1]);  
        Integer mark3 = Integer.valueOf(args[2]);  
  
        // Autounboxing happens during calculation  
        int total = mark1 + mark2 + mark3;  
        double average = total / 3.0;  
  
        System.out.println("Total Marks = " + total);  
        System.out.println("Average Marks = " + average);  
    }  
}
```

Input

Run the program as:

```
java MarksDemo 80 75 90
```

Output

```
Total Marks = 245  
Average Marks = 81.66666666666667
```